Lead Time of Dengue Outbreak Alerts in the 2024 Epidemic Year for Seen and Unseen Brazilian Municipalities under a Weekly Alert Capacity

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***Abstract*—Brazil recorded its largest dengue epidemic in 2024, and health secretariats can act on only a limited number of municipalities each week. An alert is therefore useful only when it arrives early and when most alerted places truly enter an outbreak. Existing studies mostly score case counts for states or a few cities. We build a weekly panel of 400 Brazilian municipalities from the open OpenDengue database and define an outbreak relative to the own history of each municipality. Ten classifiers spanning linear, neighbour-based, tree-based, boosted and neural models are trained on years up to 2022 and tested on the 2024 epidemic, and alert lead times are studied for calm municipalities, including municipalities never seen in training, under a fixed share of weekly alerts. The models rank municipalities well, yet under the 75th-percentile outbreak definition, their weekly top-10% alert lists are not clearly better than a simple rule that selects the places closest to their outbreak threshold. Precision stays nearly constant as the lead grows while recall falls, and the number of weeks for which an alert can be trusted depends strongly on the precision required. The contribution is a reproducible, capacity-aware benchmark with bootstrap uncertainty on public data.**

***Index Terms*—dengue, early warning, outbreak alert, forward-in-time validation, spatial hold-out, alert capacity, public health surveillance**

I. Introduction

Dengue is a mosquito-borne disease, and Brazil lived through an exceptional epidemic season in 2024: in the data used here the year holds about three times as many reported cases as the previous peak year. Health secretariats respond to such a season with a fixed workforce, so they can visit, spray or communicate with only a limited number of municipalities each week. An alert is therefore useful when two conditions hold at once, because it must arrive early enough to act and most of the places it names must really enter an outbreak. A secretariat that receives a list of one hundred high-risk municipalities when it can serve ten gains little from a model that is accurate on average. This paper studies alerts from that operational point of view rather than as a case-count forecasting exercise.

Research on Brazilian dengue forecasting is active. A national forecasting sprint evaluated probabilistic forecasts of weekly cases for five states over the 2023 and 2024 seasons [1]. Chen and Moraga forecast weekly cases for all 27 states with long short-term memory (LSTM) networks that use lagged climate variables and spatial effects [2], and later built an LSTM model with mobility-adjusted imported cases for ten large cities, which they evaluated for both case counts and outbreak detection [3]. Villela studied the risk of a high-incidence season in Brazilian municipalities with path signatures and lasso regression, and reports a sensitivity of about 75% [4]. These studies advance forecasting, but they score count accuracy for states or a few cities, or they predict a seasonal label, and among the studies we found none asks how many weeks ahead a municipal alert stays reliable when a model fixed before the epidemic year is applied to the epidemic itself.

We therefore frame the task as an early-warning classification problem at the municipal level. A municipality is in an outbreak in a given week when its reported cases exceed the 75th percentile of its own weekly counts up to 2022. Models are trained on years up to 2022, their hyperparameters are fixed in advance, and they are evaluated on the 2024 epidemic without any information from that year. We study municipalities that are still calm when the alert is issued, because alerting a place that is already in an outbreak adds little, and we limit the alerts to a fixed share of municipalities each week. Three research questions are answered. RQ1 asks how cases are distributed over years, months and states, and how exceptional 2024 is. RQ2 asks which of ten classifiers spanning linear, neighbour-based, tree-based, boosted and neural models best predicts an outbreak four weeks ahead. RQ3 asks how far ahead future threshold exceedance can be flagged for municipalities that are calm at alert time, whether results hold for municipalities never seen in training, and what precision and recall a fixed weekly alert share provides.

Our contributions are threefold. First, we provide a reproducible, capacity-aware, forward-in-time benchmark on public data, with lead times from one to eight weeks and a spatial hold-out of municipalities. Second, under the 75th-percentile outbreak definition, we show with bootstrap intervals over municipalities that learned models rank municipalities better than a one-line rule, but their weekly top-10% alert lists are

not clearly better than the rule. Third, we show that precision stays almost constant as the lead grows while the lift over the base rate roughly halves, and that the number of weeks for which an alert can be trusted depends strongly on the precision that is required. The remainder of the paper reviews related work, describes the data and method, reports the results, and discusses implications and limitations before concluding.

II. Related Work

These works differ in the unit they predict, which is a state, a city or a season, and in the target, which is a case count or a seasonal label. We focus instead on future threshold exceedance at the municipal level and evaluate how far ahead an alert remains useful under a fixed weekly alert capacity.

Gradient-boosted trees such as LightGBM and XGBoost and random forests are strong defaults for tabular data [7]–[9], and Grinsztajn et al. report that tree-based models still outperform deep learning on medium-sized tabular data [10]. We therefore compare ten classifiers spanning linear, neighbour-based, tree-based, boosted and neural models, implemented with scikit-learn [11] and using features built in SQL on DuckDB [12]. A comparison among many models is only informative when a simple reference is included, so we add a one-line rule that ranks municipalities by how close their current cases are to their own outbreak threshold. Kapoor and Narayanan show that data leakage is a frequent cause of over-optimistic results in machine-learning-based science [13]. We guard against it by building every feature from the current and earlier weeks only, by freezing each threshold at its 2022 value, and by checking that both the feature and label windows of the training rows end before the test year.

Roberts et al. argue that random cross-validation overstates performance for data with temporal or spatial structure and recommend blocked validation [14]. We follow this advice with a forward split by year and with a spatial hold-out of municipalities. Van Calster et al. stress that probabilities which are not calibrated can lead to poor decisions [15], so we report the expected calibration error for the calibration year 2023 and for the test year 2024. Demšar recommends careful statistical comparison of classifiers [16], but the tests he describes need many datasets, whereas we have one test year. We therefore quantify uncertainty by resampling municipalities with the bootstrap and by comparing every model with the simple rule on the same resamples. This keeps the claims of the paper within what a single epidemic year can support.

III. Data and Preprocessing

The raw data come from OpenDengue version 1.3 [6]. Its Brazilian part has 1,895,538 municipality-week counts for 5,517 municipalities, from the end of 2013 to the end of 2024. We keep the 400 municipalities with the most cases up to 2021 and build a complete weekly grid for them, which gives 229,600 rows over 574 weeks. Weeks that are absent from the source are set to zero cases and flagged, and 15,569 rows (6.8%) are of this kind. Table I shows that these filled weeks are almost absent until 2021 but account for 28.4% of the rows in 2022, 23.9% in 2023 and 15.8% in 2024, so 14,211 of the 15,569 filled weeks fall in the last three years. We keep them, because dropping weeks would break the lags, and we report their effect on the results in Section V.

TABLE I
Reported cases, zero-filled weeks and outbreak share by year for the 400-municipality panel.

Year	Cases (10^3)	Zero-filled (%)	Outbreak share (%)
2015	1,288	0.2	42.3
2016	1,151	0.2	26.2
2017	183	1.2	6.1
2018	201	0.5	8.5
2019	1,213	0.0	36.6
2020	623	1.0	22.6
2021	369	2.8	13.6
2022	788	28.4	25.5
2023	943	23.9	33.7
2024	**4,013**	15.8	**51.0**

An outbreak is defined for each municipality separately. The threshold is the 75th percentile of its weekly counts in the years up to 2022, with a floor of five cases, and the threshold is frozen from then on. For at least a quarter of the municipalities the threshold equals the floor, and the median threshold is about eight cases per week, so an outbreak here means a clear rise over the usual level of a place and not a large absolute number. The target y_h equals one when the cases h weeks later exceed the threshold, for $h = 1, \ldots, 8$, and the main horizon is $h = 4$. Eleven features use only the current and earlier weeks: log-transformed cases at lags 0 to 3 and 52, the four-week mean of log cases, the weekly growth of log cases, the average log cases of the state in the same week, the log threshold, and the sine and cosine of the week of the year.

The split follows time. Years 2015 to 2022 are used for training, 2023 only to assess calibration, and 2024 for testing, which gives 166,800, 21,200 and 17,200 rows after the lags and the eight-week look-ahead are applied. The test weeks run from January to the end of October 2024. The share of weeks that are outbreak weeks four weeks ahead is 22.6% in training, 33.7% in 2023 and 51.0% in 2024, so the test year is much more severe than anything the models have seen. Six of the eleven features have a test mean that lies more than half a training standard deviation away from the training mean, and the largest shift is that of the state-level cases. Municipalities that are already above their threshold are excluded from the alert analysis of RQ3, which leaves 7,913 calm test rows with an outbreak rate of 12.1% at four weeks.

IV. Method

Figure 1 shows the research pipeline. Data cleaning, SQL queries and exploratory analysis are done in the first notebook, and the models and the alert protocol in the second notebook. All code uses fixed random seeds, and the second notebook checks the processed file against a hash written by the first.

A. Classifiers and RQ2 protocol

Ten classifiers spanning five model types are compared: logistic regression and a linear support vector machine, k-

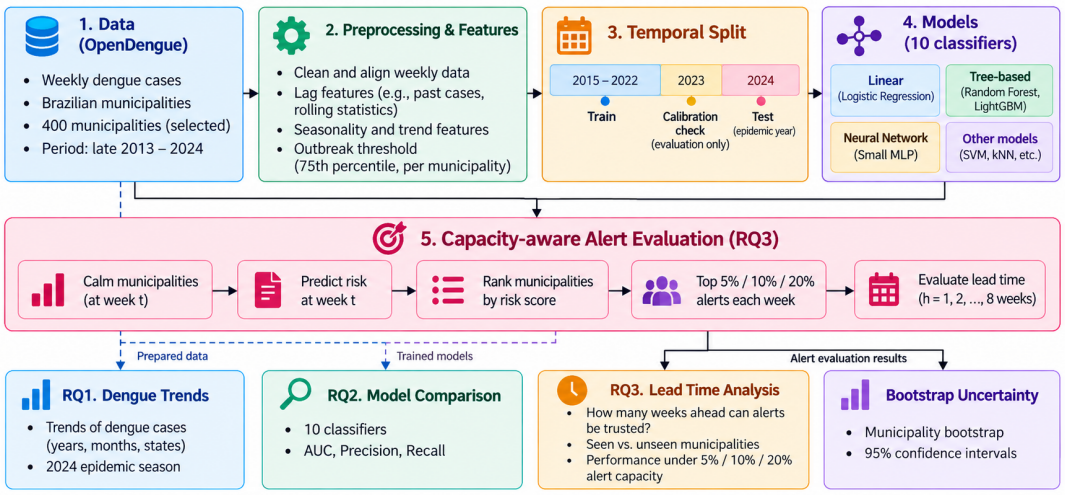


Fig. 1. Research pipeline for dengue early warning evaluation and capacity aware alerting.

nearest neighbours, a decision tree, random forest, extra trees, histogram gradient boosting, LightGBM, XGBoost, and a small multilayer perceptron (MLP) with two hidden layers of 64 and 32 units. Models that depend on scale are preceded by standardisation. Hyperparameters are fixed in advance and are not tuned on the test year: for example, the forests use 300 trees with a minimum leaf size of five, LightGBM and XGBoost use 300 trees with learning rate 0.05, and the MLP stops early on a validation part of the training rows. Training uses a random sample of at most 60,000 rows from the training years to keep ten models affordable, and each model is run with five seeds. We report ROC-AUC and PR-AUC for ranking quality and the Brier score for probability quality, using the logistic transform of the decision value for the linear SVM, whose Brier score is therefore not calibrated.

B. Alert protocol for RQ3

For each lead time $h = 1, \ldots, 8$ four representative models are trained again on the 46,468 calm rows of the training sample (rows whose current cases are not above the threshold): logistic regression, random forest, LightGBM and the MLP. Each week the calm municipalities are ranked by predicted probability and the top share is alerted, with 10% as the main share and 5% and 20% as alternatives. We report the precision and recall of this weekly list, the lift over the base rate of the same lead time, and the expected calibration error with ten bins. A simple rule serves as reference and ranks municipalities by the difference between the log current cases and the log threshold. For municipalities never seen in training, a random 30% of the municipalities (120) are removed from the training pool, a sample of 60,000 calm rows is drawn from the remaining municipalities for each of the ten models, and the models are then evaluated separately on seen and unseen municipalities.

C. Uncertainty and trusted lead time

Seeds only change the initialisation of some models and do not reflect the sampling of municipalities. We therefore draw 1,000 bootstrap samples of municipalities with replacement. In each replicate, we rerank the sampled municipality-weeks separately within each week and recompute the top-10% alert list, rather than keeping the original alert list fixed. Ties at the alert cutoff receive fractional weights so that the total alert weight equals the available number of slots. We then recompute ROC-AUC, precision and recall for each model and the threshold-gap rule. Because all methods use the same bootstrap resamples, we report paired differences between each learned model and the rule. A difference is considered clear when its 95% bootstrap interval excludes zero. The trusted lead time is the longest uninterrupted run from week one for which precision stays above a required level. We test precision thresholds of 0.30, 0.35 and 0.40; this is a sensitivity analysis of alert reliability, not a guarantee of outbreak onset prediction.

V. Results

A. RQ1: distribution of cases

The 400 municipalities report 4.01 million cases in 2024, 4.26 times the 2023 value and 3.1 times the previous peak year, 2015. Cases are concentrated early in the year: March and April together hold 43% of all cases over the period and February to May hold 72%. Several states are far above their own history in 2024. São Paulo reports 1.75 million cases, 8.2 times its annual mean for 2015 to 2023, while Amapá and Santa Catarina exceed ten times their mean. Among the municipality-weeks of the panel, 51.0% are outbreak weeks four weeks ahead in 2024, compared with 6.1% to 42.3% in 2015 to 2023.

B. RQ2: ten classifiers at four weeks

All ten classifiers separate outbreak from non-outbreak weeks well when every test week is used (Table II). Nine

TABLE II
TEN CLASSIFIERS AT A FOUR-WEEK HORIZON, TEST YEAR 2024, MEAN OVER FIVE SEEDS (STANDARD DEVIATION OF ROC-AUC IN BRACKETS). BEST VALUE IN EACH COLUMN IN BOLD.

Model	ROC-AUC	PR-AUC	Brier	Train (s)
Logistic regression	0.9532 (0.0000)	0.9594	0.0895	**0.05**
Linear SVM	0.9530 (0.0000)	0.9591	0.1154	0.09
k-NN	0.9430 (0.0000)	0.9439	0.0950	0.09
Decision tree	0.9092 (0.0003)	0.9048	0.1134	0.30
Random forest	0.9541 (0.0001)	0.9608	0.0874	20.29
Extra trees	0.9536 (0.0001)	0.9602	0.0901	8.78
HistGradientBoosting	0.9557 (0.0007)	0.9615	0.0857	0.85
LightGBM	0.9564 (0.0001)	0.9623	0.0849	1.53
XGBoost	0.9565 (0.0001)	0.9623	0.0850	1.56
MLP	**0.9572** (0.0016)	**0.9626**	**0.0845**	3.23

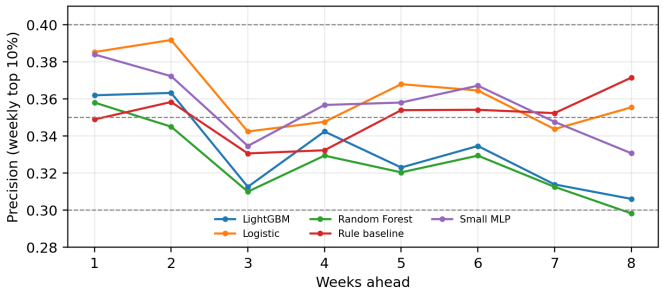


Fig. 2. Precision of weekly top-10% alerts by lead time for calm municipalities. Dashed lines show required precision levels.

of them reach a ROC-AUC between 0.943 and 0.957, the MLP is the highest at 0.957, and only the decision tree is clearly lower at 0.909. The eight best models differ by less than 0.005, and their seed standard deviation is at most 0.0016, so their order should not be over-interpreted. Models without randomness in training, such as logistic regression, show a standard deviation of zero because the training sample is fixed. These pooled scores are high partly because 51% of the test weeks are outbreak weeks and a municipality that is already in an outbreak usually stays there, which is why RQ3 restricts the analysis to calm municipalities.

C. RQ3: lead time for calm municipalities

When only calm municipalities are scored, the task is much harder, with ROC-AUC between 0.82 and 0.89 depending on the model and lead time (not shown in a figure). At one week the best models reach 0.89 and at eight weeks the MLP is still at 0.84, while logistic regression falls to 0.82. Precision of the weekly top 10% stays between 0.30 and 0.39 at every lead time, but the base rate of future outbreak events among calm municipalities rises from 8.0% at one week to 15.5% at eight weeks. The lift over the base rate therefore falls from 4.3–4.8 at one week to 1.9–2.4 at eight weeks, and recall falls from about 0.47 to between 0.19 and 0.24. The threshold-gap rule follows the learned models closely, and at eight weeks its precision of 0.374 is higher than that of every model.

How many weeks an alert can be trusted depends on the precision that is required (Table III). With a requirement of 0.30, the learned models meet the threshold continuously for

TABLE III
TRUSTED LEAD TIME IN WEEKS (UNINTERRUPTED FROM WEEK ONE) FOR THREE REQUIRED PRECISION LEVELS. BEST VALUE IN EACH COLUMN IN BOLD.

Model	≥0.30	≥0.35	≥0.40
Logistic regression	**8**	**2**	**0**
Random forest	7	1	**0**
LightGBM	**8**	**2**	**0**
MLP	**8**	**2**	**0**
Threshold-gap rule	**8**	0	**0**

TABLE IV
CALM MUNICIPALITIES SEEN AND UNSEEN IN TRAINING, TEST YEAR 2024, FOUR WEEKS AHEAD. BEST VALUE IN EACH COLUMN IN BOLD.

	ROC-AUC		Precision, top 10%	
Model	Seen	Unseen	Seen	Unseen
Logistic regression	0.852	0.845	0.348	0.343
Linear SVM	0.848	0.841	0.343	0.333
k-NN	0.789	0.796	0.249	0.238
Decision tree	0.756	0.726	0.234	0.238
Random forest	0.830	0.825	0.308	0.267
Extra trees	0.842	0.834	0.344	0.281
HistGradientBoosting	0.849	0.846	0.335	0.295
LightGBM	0.847	0.843	0.344	0.286
XGBoost	0.848	0.843	0.333	0.286
MLP	**0.863**	**0.855**	**0.365**	**0.357**

seven to eight weeks. At 0.35, the continuous lead time falls to one week for random forest and two weeks for logistic regression, LightGBM and MLP; no model qualifies at 0.40. Because the precision curves are flat and cross the 0.35 line several times, a small change in the level moves the answer from zero to eight weeks if interruptions are ignored, which is why we count weeks without interruption and report a range instead of a single number.

D. RQ3: municipalities never seen in training

For nine of the ten models the ROC-AUC on unseen municipalities is within 0.01 of the value on seen ones (Table IV). k-NN is slightly better on unseen municipalities, and the decision tree shows the largest gap at 0.031. Precision of the weekly top 10% tells a different story for tree ensembles, which lose four to six points on unseen municipalities, for example LightGBM from 0.344 to 0.286 and extra trees from 0.344 to 0.281. Logistic regression and the MLP lose less than one point, while the linear SVM loses about one point. The unseen group has 2,137 calm test rows from 120 municipalities and only about five alerts per week, so these precision values are noisy and should be read as indications.

E. RQ3: alert capacity, calibration and filled weeks

Table V gives the effect of the weekly alert share for the MLP, the best model of RQ2, trained on calm municipalities. Alerting 5% of municipalities gives a precision of 0.40 and captures 16% of future outbreak events, while alerting 20% captures 48% at a precision of 0.29, and the base rate is 12.1%. The expected calibration error of the MLP trained on all rows is 0.034 for 2023 and 0.037 for 2024, so calibration holds although the outbreak share rises from 33.7% to 51.0%. When

TABLE V
EFFECT OF THE WEEKLY ALERT SHARE FOR CALM MUNICIPALITIES (MLP, FOUR WEEKS AHEAD, BASE RATE 12.1%).

Alert share	Precision	Recall	Lift
5%	**0.402**	0.156	**3.3**
10%	0.357	0.287	2.9
20%	0.293	**0.479**	2.4

the zero-filled test weeks are removed, the ROC-AUC of one MLP run (seed 4; five-seed mean 0.957) changes from 0.959 to 0.948, indicating that the filled weeks increase the pooled score by about 0.01.

F. Uncertainty over municipalities

The reranked bootstrap estimates show that the learned models have higher ROC-AUC than the threshold-gap rule: the paired differences range from 0.068 for random forest to 0.094 for the MLP, and all four 95% intervals exclude zero (Table VI). The result is different for the operational alert list. The paired precision differences range from -0.003 to $+0.024$, while the recall differences range from -0.002 to $+0.020$; all of these intervals include zero. Thus, the models rank calm municipalities better overall, but this analysis does not establish that their weekly top-10% alert lists outperform the simple rule.

VI. DISCUSSION

A. Implications for alert operators

For a secretariat that can serve about one in ten calm municipalities each week, the list is a prioritisation aid and not a verdict. About one alert in three is followed by threshold exceedance four weeks later, which is nearly three times the base rate, and the list catches about 29% of all such future events. Halving the list to 5% raises precision to 0.40 but leaves 84% of starts unannounced, and doubling it to 20% catches 48% at the price of about seven false alerts in every ten. Reaching further ahead mainly costs coverage, because recall falls to roughly a fifth at eight weeks while precision is almost unchanged. These numbers describe a forward test on one severe season, so they indicate what an operator can expect and not a guarantee for another year.

B. Why a simple rule is hard to beat

The strongest single feature, the current log cases, has a rank correlation of 0.57 with the four-week target in the training data, and the rule is built from it and from the threshold. Municipalities that are near their threshold are the ones most likely to cross it soon, and they also dominate the top of any sensible list, which leaves little room for a model to add value there. The learned models do add value over the whole ranking, as the difference in ROC-AUC shows, but the weekly budget only looks at the top slice. A plausible reading, which we did not test, is that the models help mostly in the middle of the ranking, where operators do not look under a tight budget. The calibration result is also encouraging, because a probability that stays reliable in a year with a very different outbreak share is easier to use in a decision than a bare ranking.

C. Limitations

The data have an important weakness. Weeks that are missing in the source are treated as zero cases, and their share jumps from below 3% until 2021 to 28.4% in 2022, 23.9% in 2023 and 15.8% in 2024. These may be weeks without a report and not weeks without cases, and the outbreak thresholds, which use the years up to 2022, may be affected in a way we did not measure. We could only check the effect on ROC-AUC: for the same MLP run (seed 4; five-seed mean 0.957), ROC-AUC fell from 0.959 to 0.948 when the zero-filled weeks were removed. We did not repeat this check for precision and recall. The median threshold is about eight cases, so many outbreak labels describe modest increases in small places.

The design has limits as well. Only one test year and one country are used, and the 2024 outbreak share of 51% is far above the 34% of 2023, so the results do not automatically transfer to other years or countries.

The outbreak threshold is based on the 75th percentile of each municipality's own history. Sensitivity analysis using the 60th-percentile definition found that LightGBM and the MLP improved top-10% alert precision over the threshold-gap rule by 0.039 and 0.037, respectively, with paired 95% bootstrap intervals excluding zero. This indicates that conclusions about alert-list performance depend on the outbreak definition. The lead-time curves are computed for four representative models and the rule, while all ten models are compared only in RQ2, with training samples of at most 60,000 rows. The panel contains the 400 municipalities with the most cases, so small municipalities are not represented.

Finally, the inference is limited. Seeds do not capture the sampling of municipalities, so we added a bootstrap over the 375 municipalities represented in the calm test set and reranked the top-10% alert list within each sampled week. The models are not retrained inside the bootstrap, so the intervals capture uncertainty from municipality sampling and alert-list composition, but not uncertainty from model fitting. The trusted lead time depends strongly on the required precision, and we therefore report a range instead of one number. No climate or mobility variable is used, nothing in the study is causal, and the evaluation is retrospective, so it has not been tested in real operation. Future work should add weather inputs, test other seasons and countries, vary the outbreak definition, and retrain models inside the bootstrap.

VII. CONCLUSION

We studied how early and how reliably future municipal dengue threshold exceedances can be flagged in the 2024 Brazilian epidemic with models fixed before the season. For RQ1, 2024 holds about three times the cases of the previous peak year; cases are concentrated in March and April across all years, and half of all municipality-weeks are outbreak weeks. For RQ2, nine of ten classifiers reach a ROC-AUC of 0.943 to 0.957 at four weeks, and the differences among the best eight

TABLE VI

MUNICIPALITY-BOOTSTRAP ESTIMATES AND PAIRED DIFFERENCES TO THE THRESHOLD-GAP RULE (1,000 RESAMPLES; TOP 10% RERANKED WITHIN EACH WEEK). ENTRIES ARE ESTIMATE [95% CI]; Δ IS MODEL MINUS RULE.

	Metric estimate [95% CI]			Δ vs. rule [95% CI]		
Model	AUC	Precision	Recall	Δ AUC	Δ Precision	Δ Recall
Threshold-gap rule	0.769 [0.746, 0.790]	0.332 [0.294, 0.370]	0.267 [0.237, 0.298]	–	–	–
Logistic regression	0.852 [0.830, 0.870]	0.348 [0.316, 0.390]	0.280 [0.253, 0.317]	**+0.083 [0.063, 0.104]**	+0.015 [−0.010, 0.051]	+0.012 [−0.008, 0.041]
Random forest	0.837 [0.816, 0.856]	0.329 [0.295, 0.369]	0.265 [0.240, 0.296]	**+0.068 [0.047, 0.090]**	−0.003 [−0.033, 0.034]	−0.002 [−0.028, 0.026]
LightGBM	0.847 [0.828, 0.866]	0.342 [0.303, 0.378]	0.276 [0.246, 0.306]	**+0.078 [0.058, 0.098]**	+0.010 [−0.022, 0.040]	+0.008 [−0.018, 0.033]
Small MLP	0.862 [0.843, 0.880]	0.357 [0.315, 0.396]	0.287 [0.255, 0.317]	**+0.094 [0.075, 0.113]**	+0.024 [−0.003, 0.050]	+0.020 [−0.002, 0.040]

are too small to rank them. For RQ3, calm municipalities can be ranked well up to eight weeks ahead, precision of a weekly top 10% stays between 0.30 and 0.39 while recall falls from 0.47 to about 0.21, and the trusted lead time ranges from none to eight weeks depending on the required precision.

Under the 75th-percentile outbreak definition, learned models outperform a one-line rule in ROC-AUC but do not clearly improve the weekly top-10% alert list. Tree ensembles lose precision on municipalities never seen in training, while calibration remains similar from 2023 to 2024. The benchmark, with its alert budget and municipality bootstrap, gives public-health analysts a realistic account of what such alerts can deliver.

VIII. DATA AND CODE AVAILABILITY

The data used in this study are from the OpenDengue version 1.3 release [6]. The notebooks, SQL queries, and derived tables are available at https://github.com/Winnnnne/Dengue-Outbreak-Lead-Time/tree/main.

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